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Design and Performance Evaluation of a Resilient Smart Campus Network using Modern Layered Protocols

Course Code & Name: CSA0704 - Computer Networks

Team: Student A (Lead/Topology), Student B (Routing & Network Layer), Student C (Transport & Analysis), Student D (Application + Documentation)

1. Problem Understanding & Formulation

A large multi-campus university network is being redesigned to support high-density Wi-Fi, IoT energy-monitoring sensors, online teaching platforms, research data transfers, and administrative services. The current network suffers from wireless collisions, inefficient IP allocation, slow routing convergence during link failures, TCP congestion collapse during peak lecture hours, and unreliable application services (DNS delays, email failures, unsecured file transfer).

Entities: multi-building campus LAN/WLAN, 5000+ concurrent wireless users, 500 IoT devices, research data links up to 1 Gbps between buildings, DNS/HTTP/mail/file-transfer services.

Traffic types: high-density Wi-Fi client traffic, IoT sensor telemetry, bulk research data transfer, lecture-time bursty TCP traffic, DNS/email/administrative traffic.

Failure modes considered: link failure requiring routing re-convergence, wireless collision/contention under high density, TCP congestion collapse during peak load, DNS resolution delay, unreliable/insecure file transfer.

Assumptions & constraints: hierarchical IPv4 addressing (optional IPv6), average wireless packet loss below 1%, local DNS resolution under 50 ms, TCP goodput recovery within 2 RTT after congestion, routing convergence under 5 seconds for a single link failure.

2. Application of Course Knowledge

2.1 Data Link Layer

Error control: CRC is used for frame-level error detection across Ethernet and Wi-Fi links; Hamming code is discussed as an alternative for single-bit error correction in low-rate IoT sensor links.

Medium access control: CSMA/CA for legacy Wi-Fi contention, and OFDMA (Wi-Fi 6 / 802.11ax) for scheduled, collision-reduced access in high-density zones.

LAN technologies: Gigabit Ethernet backbone between buildings, Wi-Fi 6 (802.11ax) access points for classrooms/hostels, Bluetooth Low Energy for IoT sensor connectivity.

2.2 Network Layer

Hierarchical IPv4 addressing is planned per building/VLAN block to keep routing tables compact and allow efficient subnetting. Subnetting is calculated per department/building based on host-count requirements (worked subnetting table to be inserted by Student B from the Packet Tracer configuration).

Routing: OSPF (link-state) is proposed over RIP (distance-vector) for faster convergence (target under 5 seconds) and better scalability across multiple buildings; a brief DV vs LS comparison table is included below.

Aspect	Distance-Vector (RIP)	Link-State (OSPF)
Convergence speed	Slow, prone to loops	Fast, loop-free
Scalability	Poor for large campus	Good, hierarchical areas
Overhead	Low CPU, periodic updates	Higher CPU, event-driven

ICMP is used for diagnostics (ping/traceroute) to verify end-to-end reachability and measure convergence after simulated link failure.

SDN/OpenFlow concept: at least one campus segment (e.g., the IoT sensor VLAN) is proposed to be centrally controlled, separating the control plane from switching hardware to allow centralized, policy-based traffic management and easier fault isolation.

2.3 Transport Layer

TCP is used for reliable services (HTTP, email, file transfer); UDP for latency-tolerant/real-time IoT telemetry; QUIC is discussed as a modern alternative for encrypted, low-latency web traffic. Congestion control (TCP Reno/Cubic behaviour) is analysed for peak lecture-hour load, with QoS via priority queuing for latency-sensitive traffic. A simple M/M/1 queuing model is used to estimate average queuing delay at a bottleneck link.

2.4 Application Layer

A hierarchical DNS setup resolves campus hostnames locally (target under 50 ms); HTTP/HTTPS with TLS 1.3 secures web services; SMTP/IMAP handles campus mail; SFTP/FTPS is used instead of plain FTP for secure file transfer; SNMP provides basic device monitoring. Security implications (TLS handshake overhead, DNS spoofing risk, mail spoofing) are discussed alongside mitigations.

3. Solution / Design / Methodology

Proposed topology: a hybrid star-of-stars design - each building has a star-wired LAN (access switches to a building distribution switch), buildings are interconnected via a mesh/backbone core for redundancy, and Wi-Fi 6 APs provide wireless coverage per building. IoT sensors connect through a dedicated VLAN with SDN-based policy control.

Two alternative designs were considered: (1) a flat single-VLAN campus network - simple but does not scale and mixes traffic types, increasing congestion and security risk; (2) the chosen hierarchical, VLAN-segmented, OSPF-routed hybrid design - segments traffic by building/function, supports redundancy through the mesh backbone, and meets the sub-5-second convergence requirement. The hierarchical design was selected for scalability, fault isolation, and alignment with the performance constraints given in the problem statement.

[Insert Packet Tracer topology diagram and addressing plan table here.]

4. Implementation using Modern Tools

Packet Tracer is used to build the multi-building topology: core/distribution/access switches, routers running OSPF, Wi-Fi 6 access points, DHCP/DNS/HTTP servers, and end devices. Wireshark captures are taken at key points - wireless association, ARP, ICMP, TCP three-way handshake, DNS query/response, and HTTP/TLS handshake - to verify correct protocol behaviour.

[Insert screenshots: end-to-end ping success, routing table, wireless client association, DNS/HTTP response, Wireshark capture filters used.]

5. Results, Validation & Analysis

[Insert quantitative results table: throughput, average delay, packet loss %, DNS resolution latency, routing convergence time, TCP retransmission count - obtained from Packet Tracer Simulation Mode and/or Wireshark statistics.]

Trade-off discussion: the hierarchical OSPF design trades slightly higher configuration complexity for significantly faster convergence and better scalability compared to the flat/RIP alternative, which is justified given the campus's scale (5000+ wireless users, 500 IoT devices) and the sub-5-second convergence requirement.

6. Broader Considerations & Conclusion

Sustainability: Wi-Fi 6 Target Wake Time reduces IoT device power consumption; hierarchical routing reduces unnecessary broadcast traffic, lowering overall network energy use. Security/privacy: TLS 1.3 and SFTP protect data in transit; SDN-based policy control on the IoT segment limits lateral movement in case of a compromised sensor. Accessibility: reliable, low-latency DNS/HTTP services support remote learners. This design aligns with SDG 4 (Quality Education) through reliable online learning infrastructure, SDG 9 (Industry, Innovation and Infrastructure) through modern SDN/Wi-Fi 6 adoption, and SDG 11 (Sustainable Cities and Communities) through energy-efficient network operation.

Conclusion: the proposed hierarchical, VLAN-segmented, OSPF-routed campus network with an SDN-controlled IoT segment addresses the stated challenges of collisions, inefficient addressing, slow convergence, congestion, and unreliable application services. Limitations include the simulated (rather than physical) evaluation environment and time constraints on full-scale load testing; future work would extend queuing analysis and add IPv6 dual-stack support.

7. Team Reflection

Student A: [2-3 lines - what was learned building the topology and addressing plan.]

Student B: [2-3 lines - what was learned on subnetting/routing/SDN.]

Student C: [2-3 lines - what was learned analysing TCP/UDP/congestion with Wireshark.]

Student D: [2-3 lines - what was learned on application services and GitHub/report coordination.]

8. References

- IEEE 802.11ax (Wi-Fi 6) Standard
- Relevant RFC documents (DNS, HTTP, TCP, OSPF)
- Cisco Packet Tracer Documentation
- Wireshark User Guide

Team Roles

Member	Primary Responsibility	Key Tools
Student A (Lead / Topology)	Overall network design, Packet Tracer topology, addressing plan, VLANs, wireless APs	Packet Tracer
Student B (Routing & Network Layer)	IP subnetting tables, routing protocols (OSPF/EIGRP), SDN concept, ICMP diagnostics	Packet Tracer + config

Student C (Transport & Analysis)	TCP/UDP/QUIC discussion, congestion & QoS analysis, Wireshark captures, performance metrics	Wireshark + Simulation Mode
Student D (Application + Documentation)	DNS, HTTP/HTTPS+TLS, SMTP/FTP, SNMP; GitHub repo setup, report compilation, reflection	Wireshark + GitHub + Report

Team Contribution Statement

- Student A: Topology design, Packet Tracer model, addressing overview, wireless configuration
- Student B: Subnetting tables, routing configuration & convergence tests, SDN concept note
- Student C: Wireshark captures & analysis, transport-layer (TCP/UDP/congestion) evaluation, performance metrics
- Student D: Application services (DNS/HTTP/TLS), GitHub repository management, report compilation & team reflection coordination

All members jointly reviewed results, trade-offs and final conclusions.